

Roll No.

31476

**B. Pharm. 3rd Semester
(Regular/Re-Appear/Improvement) (Mercy
Chance) Examination – December, 2023**

PHARMACEUTICAL ORGANIC CHEMISTRY-II

Paper : BP301T

Time : Three Hours /

/ Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt *all* questions from Section-A, each question carries 2 marks. Attempt any *two* questions from Section-B, each question carries 10 marks. Attempt any seven questions from Section-C, each question carries 5 marks.

SECTION – A

1. Write brief notes on the following :

$2 \times 10 = 20$

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- (a) Resonance in benzene
- (b) Acidity of phenols
- (c) Orientation of mono-substituted benzene towards electrophilic substitution reactions
- (d) Effect of substituents in benzene on reactivity of benzene

1885 12/05
(c) Drying of oils

(d) Iodine value

(g) Structure of Naphthalene

(h) Medicinal uses of Anthracene

(i) Hydrolysis of oils and fats

(j) Uses of saccharine

SECTION – B

2. Explain the effects of various substituents on reactivity and orientation of mono-substituted benzene compounds towards electrophilic substitution reaction.

3. What do you understand by basicity of amines ? Write effects of various substituents on the basicity of amines. 10
4. What do you understand by polynuclear hydrocarbons ? Write structure and medicinal uses of at least *four* polynuclear hydrocarbon compounds. 10

SECTION - C

5. Write structure and uses of DDT and saccharine. 5
6. Explain orbital structure of benzene and aromatic character of benzene. 5
7. What is Baeyer's Strain theory ? Write its limitations. 5
8. Write a short note on Friedelcraft's acylation along with mechanism. 5
9. Explain principle involved in determination of acetyl value. Write its significance also. 5

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Wesley
2-12-66

10. Write synthesis and reactions of naphthalene. 5

11. Write a short note on Coulson and Moffitt's modification. 5

12. Explain aromatic character of benzene. 5

13. Write Synthetic uses of aryl diazonium salts. 5

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Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B. Pharm. 3rd Semester, 1st Sessional Examination, Oct, 2023
Paper Code –BP301T; Organic chemistry (Theory)

Duration: 1 hr 30 minutes

Total marks: 30

Question no.1 is compulsory.

(5×2=10

1. Define the following.
marks)

- a. Huckel's rule.
- b. Define fats and oils.
- c. Structure of chloramine.
- d. Give the formula of saponification value.
- e. Define sulphonation .

2. Attempt any two questions from the rest.

(2×5=10 marks)

- a. Acidity of Phenols.
- b. Structure and uses of DDT and BHC.
- c. Discuss Acid value and RM value.

3. Attempt anyone questions from the rest.

(1×10=10 marks)

- a. Discuss benzene and its derivatives, also nitration and Fridelcraft acylation.
- b. Difference between fats and oils, Synthesis and hydrolysis of triglycerides.

Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B.Pharma 3rd Semester ,2nd Sessional Examination, 2023
Paper Code -BP301T; organic Chemistry (Theory)

Duration: 1 hr 30 minutes

Total marks: 30

Question no.1 is compulsory

1 a Define acidity of phenol?

2x5=10 marks

b Give the structure of resorcinol?

c Give the structure of Phenanthrene?

d Write two uses of cresols?

e Define aromatic amines and aromatic acid?

2 Attempt any one questions from the rest

10x1=10 marks

a. what are phenols? explain the reaction of phenols and discuss the effect of substituents on acidity of phenols?

b. define polynuclear hydrocarbon and give structure and uses of triphenylmethane, phenanthrene, and naphthols?

3 Attempt any two questions from the rest

5x2=10 marks

a. What are phenols ? discuss the acidity of phenols?

b. write the structure and uses of naphthalene, anthracene and diphenylmethane?

c. what are aromatic amines? explain the basicity of aromatic amines?

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**B. Pharma. 3rd Semester Regular/Re-appear/Improvement (Mercy Chance)
Examination – December, 2023**

PHARMACEUTICAL ENGINEERING

Paper : BP304T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is ***compulsory***. Attempt any ***seven*** questions from Section-B, each carry **5** marks and ***two*** questions from Section-C, each carry **10** marks.

SECTION – A

1. Define the followings :

2 × 10 = 20

- (a) Energy losses
- (b) Size separation
- (c) Filtration

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- (d) Mixing
- (c) Drying
- (f) Conduction
- (g) Reynolds number
- (h) Equilibrium moisture content
- (i) Centrifugation
- (j) Seidtz filter

$$Q = n \frac{dx}{dr}$$

SECTION – B

2. Define flow of fluids. Write a note on various types of manometers.
3. What are mechanisms & laws governing size reduction ?
4. Explain the economy of multiple effect evaporator.
5. Write a note on Fourier's law and heat transfer mechanisms.
6. Explain basic principles and working of steam distillation.
7. Write a note applications & mechanism of drying process.

8. Write down principles, construction and working of fluidized bed dryer.
- ✓ 9. What are theories & factors influencing filtration ?
- ✓ 10. Write down principle, construction and working of Sigma blade mixer.

SECTION – C

11. Write a note on size reduction with principle, construction, working, uses, merits and demerits of Hammer mill.
12. Explain various materials of pharmaceutical plant construction.
13. Write down principle, construction, working, uses, merits and demerits of non-perforated basket centrifuge.

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari
B. Pharmacy 3rd Semester First Sessional Examination
Pharmaceutical Engineering (Paper code –BP304T)

Date- October 18, 2023

Duration- 1h 30 min

Max. Marks- 30

Q1. Question 1 is compulsory. Each part carries 2 marks.

- i. Principle of edge runner mill and end runner mill.
- ii. Application of the heat transfer.
- ii. Define conduction and convection.
- iv. Factors effecting evaporation.
- v. Define Fourier's Law.

Attempt any two out of three. Each question carries 5 marks.

Q2. Define laws governing size reduction briefly.

Q3. Diagram and working of fractional distillation?

Q4. Disadvantage and advantage of steam jacketed kettle evaporator.

Attempt any one out of two. Each question carries 10 marks.

Q5. Construction, working, merit and demerit of forced circulation evaporator.

Q6. Principle, diagram and working of Hammer mill and Ball mill.

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari

B. Pharmacy 3rd Semester 2nd Sessional Examination

Pharmaceutical Engineering (Paper code -BP304T)

Date- November 30, 2023

Duration- 1h 30 min

Max. Marks- 30

Q1. Question 1 is compulsory. Each part carries 2 marks.

- i. Principle of cyclone separator.
- ii. Define Merits of fluid energy mill.
- iii. Define Bernoulli's theorem.
- iv. Define objectives of Super Centrifuge.
- v. Define principle of Meta filter.

Attempt any two out of three. Each question carries 5 marks.

Q2. Define objectives & factors affecting size reduction.

Q3. Diagram and working of Cyclone Separator?

Q4. Disadvantage and advantage of non-perforated basket centrifuge.

Attempt any one out of two. Each question carries 10 marks.

Q5. Construction, working, merit and demerit of Perforated Basket Centrifuge.

Q6. Principle, diagram and working of Edge Runner Mill and Fluid Energy Mill.



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**B. Pharma. 3rd Sem.
(Regular/Re-appear/Improvement)
(Mercy Chance)
Examination – December, 2023**

PHARMACEUTICAL MICROBIOLOGY

Paper : BP303T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

***Note : Section-A is compulsory; each carry 2 marks.
Attempt any seven questions out of nine from
Section-B, each carry 5 marks and two questions out
of three from Section-C, each carry 10 marks.***

SECTION – A

Note : Attempt all questions.

$2 \times 10 = 20$

1. (a) Define Eukaryotes.
- (b) What is viable count ?

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- (c) Explain staining techniques for identification of bacteria.
- (d) Write the evaluation of antiseptics.
- (e) Explain the source of microbial contamination.
- (f) Differentiate between reproduction and replication.
- (g) Write the principle of sterilization.
- (h) Write a note on growth curve.
- (i) Explain designing of aseptic area.
- (j) Discuss about transformed cell cultures.

SECTION – B

Note : Attempt any *seven* questions.

$5 \times 7 = 35$

- 2. Discuss about history and scope of microbiology.
- 3. Write the classification and mode of action of disinfectants.
- 4. Discuss about laminar flow equipment.

5. Define cell culture and write general procedure for cell culture.
6. Write sterility testing of ophthalmic product according to IP.
7. Explain factor affecting microbial spoilage.
8. Write a note on biochemical test.
9. Discuss about evaluation of bactericidal and bacteriostatic.
10. Write the principle, advantage and limitation of electron microscopy.

SECTION – C

Note : Attempt any *two* questions.

2 × 10 = 20

11. Explain radiation and mechanical method of sterilization in detail.

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12. Explain principle and method of different microbiological assay.
 13. Discuss about application of cell culture in pharmaceutical industry.
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Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari

B. Pharmacy 3rd Semester 1st Sessional Examination

PHARMACEUTICAL MICROBIOLOGY (Paper code -BP 303 T)

Date- 17 October, 2023

Duration- 1h 30 min

Max. Marks- 30

Q1 is compulsory. Each part carries 2 marks.

Explain in detail

- i. Contributions of Louis Pasteur
- ii. Log phase of bacterial growth
- iii. Any two applications of UV radiations
- iv. Uses of transmission electron microscope
- v. Chemical method of sterilization

Attempt any two out of four. Each question carries 5 marks.

Q2. Write about morphological classification of bacteria.

Q3. Give the principle and applications of dark field microscopy.

Q4. Explain methyl-red test and voges-prashauer test in detail.

Q5. Write principle and procedure of Acid-fast staining.

Attempt any one out of two. Each question carries 10 marks.

Q6. Explain the various methods of sterilization.

Q7. What is microscopy. Explain different types of microscopy in detail.



Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B. Pharm. 3rd Semester, 2nd Sessional Examination, Nov. 2023
Paper Code – BP-303T; Pharmaceutical Microbiology (Theory)

Duration: 1 hr 30 minutes

Total marks: 30
(5×2=10 marks)

1. Question no. 1 is compulsory.

- a. Define the term 'morphology' in the context of fungi and provide one example of a distinctive morphological feature.
- b. List two major differences between the reproduction methods of fungi and viruses.
- c. What is the primary factor that differentiates antiseptics from disinfectants?
- d. Name one bacteriostatic and one bactericidal agent commonly used in pharmaceutical environments.
- e. Briefly explain the concept of clean area classification in pharmaceutical manufacturing.

2. Attempt any two questions from the rest.

(2×5=10 marks)

- a. Describe the classification of viruses based on their replication method and provide an example for each class.
- b. Explain the mode of action of alcohols as disinfectants and discuss two factors that influence their disinfectant activity.
- c. Outline the design considerations for an aseptic area in pharmaceutical manufacturing, focusing on contamination prevention.

3. Attempt any one questions from the rest.

(1×10=10 marks)

- a. Discuss the different sources of microbial contamination in an aseptic area, the methods used to prevent contamination, and the role of laminar flow equipment in maintaining an aseptic environment.
- b. Evaluate the importance of cell culture applications in the pharmaceutical industry and research, including the assessment of microbial contamination and spoilage of pharmaceutical products.

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**B. Pharma. 3rd Sem.
(Regular/Re-appear/Improvement) (Mercy
Chance) Examination – December-2023**

PHYSICAL PHARMACEUTICS – I

Paper : BP302T

Time : Three Hours /

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section – A is *compulsory*, each carry 2 marks. Attempt any seven questions out of nine from Section – B, each carry 5 marks. and two questions out of three from Section – C, each carry 10 marks,

SECTION – A

Note : Attempt *all* questions.

2 × 10 = 20

1. (a) Define interfacial tensions.
- (b) Write about HLB scale.

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- (c) What is refractive index ?
- (d) What are binary solutions ?
- (e) Write buffer equation.
- (f) Define physicochemical properties of drugs.
- (g) Define protein binding.
- (h) Define surfactants and write its example.
- (i) Explain optical rotation.
- (j) What is the real solution ?

SECTION – B

Note : Attempt any *seven* questions.

$5 \times 7 = 35$

- 2. Discuss about diffusion principle in biological system.
- 3. Discuss about thermodynamic treatment of stability constant.
- 4. Define buffer capacity and factor affecting it.
- 5. Write about Sorensen pH scale.
- 6. Explain Raoult's law in detail.
- 7. Write a note on state of matter.
- 8. What is dissociation constant, write its application ?
- 9. Discuss about buffered isotonic solutions.

- ✓ 10. Write the mechanism of solute solvent interaction and ideal solubility parameters.

SECTION – C

Note : Attempt any *two* questions.

$10 \times 2 = 20$

11. Define complexation and write its classification and application.
12. Discuss measurement of surface and interfacial tension in detail.
13. Define distribution law and discuss its application and limitation.
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B. Pharmacy 3rd Semester 1st Sessional Exam Oct 2023
Physical Pharmaceutics-I BP302T

Time: 1.5 Hrs

Max. Marks: 30

SECTION-A

2X5=10

Attempt all Questions. Each question carry 2 Marks.

Write a note on the following-

1. Surfactants
2. Factors affecting surface tension
3. Sublimation
4. Detergency
5. Latent heat

SECTION-B

5X2=10

Short Answer Questions. Attempt any 2 out of three.

1. Write a detailed note on HLB scale
2. Explain eutectic mixtures and relative humidity in detail.
3. Define and derive the formula to calculate spreading coefficient

SECTION-C

10X1=10

Long Answer type Question. Attempt any one out of two

1. What are the various methods for determining the surface and interfacial tension. Explain in detail about the capillary rise and DuNouy ring detachment method with derivations?
2. Write a note on- a) Polymorphism b) Amorphous and crystalline solids.